

Centre No: _____

Centre Name: _____

Code: _____ Subject: _____

Candidate Number: Alex Candidate Name: _____

Signature: _____

Do not
write in
this
marginQ₁

(a) 24

30

(b) 1-2-3-4-90
~~90~~(c) odd
even

(d) question a returns void w*h, which means the value returned will be various if ~~the~~ w and h ~~is~~ are different; question c returns a value "even" or "odd", hence if the number ~~is~~ even, the result will always be "even" instead of changing with the number.

Q₂

(a) (i) 0 to 5

(ii) 1 to 6

(iii) there is a for iteration, $i=0$ ~~and~~ $i<4$; $i++$ (iv) ~~easy to adjust and understand~~~~if the times should be changed to 5, just change $i<4$ to $i<5$~~

Reduce the lines of code, making it more simple.

Decrease the potential mistakes may be made.

Using 'Random rnd = new Random();' decreases the codes and makes it easier to understand.

(b) (i) Function Roll() Returns INTEGER
RETURN Random(1 to 6)
ENDFUNCTION

(ii) For $i \leftarrow 1$ to 5
~~Repeat~~
 OUTPUT $R_0(i)$ & " "
 NEXT i

(c) (i) 0 to 2
 circle

(ii) The $\frac{\text{cnt}}{N}$ represents the possibility that the random point lands inside a circle drawn within the square, as cnt adds 1 when the point is inside the circle.

And the 4 is the area of the square, hence $4 \times \frac{\text{cnt}}{N}$ represents the area of the circle, which is $\pi r^2 = \pi$

So the final line estimates π

(iii) The P_i is double and so do the x and y , hence it should be 4.0 in order to calculate a precise and accurate double value.

Q3 (a) Function NextTerm($n = \text{INTEGER}$) RETURNS INTEGER
 IF $n \bmod 2 = 0$ THEN
 RETURN ~~$n+1$~~ $0.5 \times n$
~~IF $n \bmod 2 \leq 20$ THEN~~
 ELSE
 RETURN $n \times 3 + 1$
 ENDIF
 ENDFUNCTION

(b) Function Steps($n = \text{INTEGER}$) RETURNS INTEGER
 DECLARE ^{Time, count} ~~Time, Count~~ INTEGER
~~Count~~
~~For $i \leftarrow 0$ to~~ Count $\leftarrow 0$
 REPEAT
 IF NextTerm > 1 THEN
 Count $\leftarrow$ Count + 1 $n \leftarrow$ NextTerm(n)
 ENDIF

(b) UNTIL NextTerm(n) = 1
RETURN Count + 1
ENDFUNCTION

(c) The time that the iteration should work isn't sure, hence the while loop should be used as the iteration will not end until the value returned by NextTerm(n) is 1.

(d) DECLARE n

INPUT n

While NextTerm(n) <> 1

OUTPUT NextTerm(n) & " "

n ← NextTerm(n)

ENDWhile

OUTPUT 1

OUTPUT Steps(~~10~~) & steps(n)

6, 8

7, 11